Jaden Majid

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Computer Science student at the University of British Columbia with experience building software systems, machine learning models, and developer tooling. Strong foundation in Python, C, Rust, and TypeScript, with a focus on reliable implementation, performance, and measurable operational impact.

Experience

Technology and Digital Intern | *Disneyland* | *Hong Kong* Sept. 2024 – July 2025

- Re-engineered database interface jobs from legacy batch processes in PowerShell, achieving 12.5x faster execution.
- Reduced data interface project cost by 95% compared to a previously submitted third party quote.
- Automated over 1,100 work-hours per year of previously manual workflows for cross-functional operations teams.
- Developed and maintained a full-stack application that digitized record-keeping and automated security incident analysis.

Digital Consulting Intern | *Deloitte* | *Hong Kong* July 2024 – Aug. 2024

- Developed Python scripts to generate synthetic test data, improving QA efficiency for marketing deployments.
- Owned a project feature from planning through client acceptance, reducing implementation cost by 80%.

Teaching Assistant | *UBC Physics Department* | *Vancouver, BC* May 2023 – Apr. 2024

- Facilitated lab sessions and lectures for 100+ students, debugging numerical simulations and grading assignments.

Personal Projects

Super Resolution GAN Image Upscaler | *Python, PyTorch* Jan. 2026 – Ongoing

- Engineered a generator-discriminator image-upscaling architecture using bottleneck residual blocks.
- Implemented a discriminator without pre-trained VGG weights, maintaining stable, non-collapsing adversarial training.
- Added a global skip connection that combines learned high-frequency residuals with a bicubic base image to improve convergence and output sharpness.
- Built modular model components from PyTorch primitives to support efficient experimentation on available training hardware.

MNIST in C | *C* Nov. 2025 – Dec. 2025

- Engineered a feed-forward neural network from first principles using only glibc, achieving 97%+ test accuracy.
- Implemented backpropagation and stochastic gradient descent from first principles with configurable layer topologies.
- Built a unit-tested linear algebra library with general matrix multiplication (GEMM) and a custom serialization format.
- Used pointers and manual memory management throughout the system, with zero leaks verified by Valgrind.

InsightUBC | *TypeScript, Vite* Sept. 2025 – Dec. 2025

- Built a full-stack analytics platform for parsing and querying the UBC PAIR dataset through a custom JSON query protocol.
- Implemented relational-algebra operations including selection, filtering, grouping, and aggregation.
- Implemented a type-safe parser from a prescribed EBNF grammar, validating abstract syntax tree nodes during query construction.

WikiScrape | *Rust, Python* Jan. 2025

- Built a concurrent web crawler using Tokio and Reqwest to map Wikipedia articles as a directed graph.
- Implemented an asynchronous worker pool that reached a burst throughput of 2,900 requests per second.

Skills

Programming/Scripting Languages: Python, C, Rust, TypeScript
Libraries/Tools: NumPy, PyTorch, Matplotlib, Vim, Linux
Languages: Native English, Intermediate Chinese, Intermediate French

Education

The University of British Columbia Sept. 2021 – Dec. 2026

B.Sc. Computer Science, Co-op Program *Vancouver, Canada*

Relevant Courses: Computer Vision, Deep Learning, Advanced Machine Learning, Software Engineering, Parallel Algorithms, Data Structures & Algorithms